Ohm’s law

antocciphysics

Name: _____ **Block:** _____ **Mastery Rating:** (teacher only) _____

LEARNING TARGET Connect voltage, current and resistance for an ohmic component.

SUCCESS CRITERIA - I can...

1. Identify the voltage across a resistor.
2. Use V=IR with appropriate units.
3. Predict changes while stating what stays fixed.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “ohmic” and “resistance” to explain your model or result. Sentence starters are optional.

ohmic: having constant resistance under the stated physical conditions

resistance: voltage divided by current for a stated operating condition

voltage: electric potential difference; energy transferred per charge

01 Warm-up: bring back an earlier idea

Move from r to 3 r away from a point charge. **Compare** field magnitude and state the source model assumption.

Word help: charge = a property responsible for electric interactions

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: V = IR; I = V/R.
Ohmic resistor under fixed physical conditions. Resistance can change with temperature.

Worked reasoning: Across a 4 ohm resistor, a 12 V supply gives I=V/R=3 A. Doubling resistance at the same voltage halves current.

Your turn: Draw a one-resistor circuit, label supply voltage and current, and mark both resistor terminals.

Sentence starter: The … in my model represents … . Its label shows … .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

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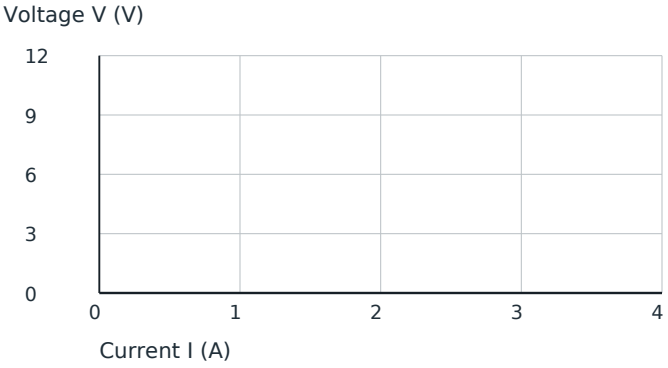
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Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Sketch a graph of V against I for a fixed ohmic resistor. What does its slope represent?

Sentence starter: The … in my model represents … . Its label shows … .

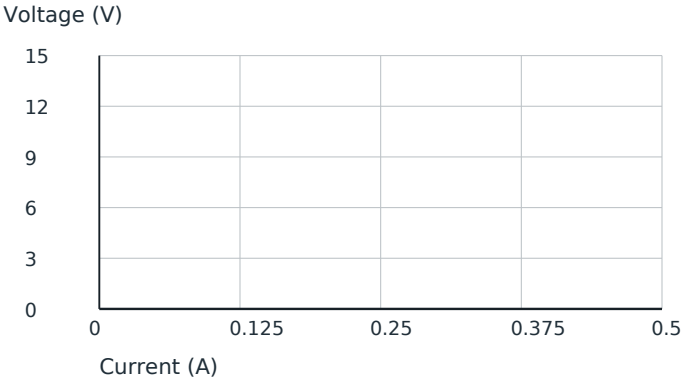


Marzano check - circle: ① ② ③

04 Apply the physics

A resistor has 12 V across its terminals and carries 0.40 A. **Find** resistance. If the voltage becomes 9 V and resistance stays constant, find the new current. **Label** both operating points on the graph.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

05 Explain your reasoning

A student doubles resistance while keeping voltage fixed and predicts twice the current. Correct the prediction.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An EM wave in vacuum has frequency 6×10^{9} Hz.
Use $c=3.00\times10^{8}$ m/s.
Find wavelength and name the spectral region.
Label a wave sketch to support the calculation or comparison.

Word help: vacuum = space with negligible matter

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

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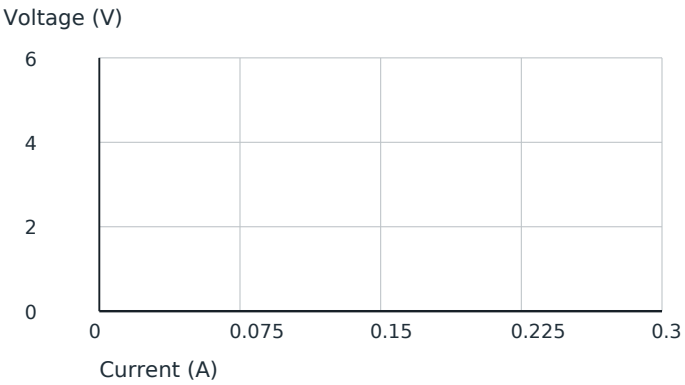
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Measured values are (I, V)=(0.10, 2.0),(0.20, 4.0),(0.30, 6.0).
Find resistance and explain what supports an ohmic model.

Sentence starter: Under the assumption …, the evidence … supports … because … .



Marzano check - circle: 1 2 3

08 STRETCH: test the model

An ohmic component has 4 V across it at 0.5 A.
Find resistance and current at 10 V.
Explain why this extrapolation requires the component to remain ohmic.

Sentence starter: Under the assumption …, the evidence … supports … because … .



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3